

Rishi Upadhyay

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EDUCATION

Ph.D, University of California, Los Angeles

Computer Science, Advised by Prof. Achuta Kadambi
2025 Amazon Fellowship, 2026 UCLA JCCC Fellowship

Sep. 2022 - June 2027

M.S., University of California, Los Angeles

Computer Science, Advised by Prof. Achuta Kadambi
Thesis: *Improving Projective Geometry in Diffusion Models*

Sep. 2022 - June 2024

B.S., University of California, Berkeley

Electrical Engineering & Computer Science

Aug. 2018 - May 2022

EXPERIENCE

Visual Machines Group, UCLA

Graduate Student Researcher

- Develop techniques to ground computer vision models in physics to enable better robustness and generalization. Recently focused on video world models, benchmarking their physical accuracy and developing novel 3D native world models that can learn and reason about physics.
- Applied these techniques across domains including improving perspective in image generation (SIGGRAPH Asia 2023), enabling sparse-view gaussian splatting with regularization (3DV 2025), and benchmarking current world models physical accuracy (in submission)

Savjani Lab, UCLA Health

Graduate Student Researcher

- Extended current techniques for respiratory motion modeling using deep learning architectures such as VAEs and Diffusion models. Models were trained and tested on an internal dataset of 5DCT scans collected from prior work.
- Demonstrated the ability of the model to generalize to new patients, overcoming a key limitation of prior work, while remaining competitive in terms of performance.

AMD Research and Development

Research Associate

Jan. 2026-Aug. 2026

- Developing 3D native object-centric world model for multi-object dynamics prediction from a single image. Model disentangles simulation and rendering, allowing for fast and accurate simulation through a lightweight physics-conditioned transformer while maintaining high quality and diverse output videos through pre-trained diffusion models.
- Demonstrated that model architecture allows it to accurately model physical laws and can provide benefits for downstream robotics tasks through both synthetic data generation and in-the-loop planning.

Meta Reality Labs (XR Eyes)

Research Scientist Intern

June 2024-Sep. 2024

- Developed end-to-end differentiable pipeline that combined deep learning and geometric eye tracking algorithms by creating a module to back-project an estimated 2D pupil ellipse into a 3D pupil.
- Demonstrated that the new pipeline was performant as a low-cost eye tracker and could be plugged into existing deep learning pipelines to improve performance across a variety of tasks.

Visual Computing Lab, UC Berkeley

Mentors: Prof. Ren Ng, Prof. Austin Roorda

- Developed a novel tool that used normalized cross-correlation and texture mapping techniques to align retinal maps that vary in cone shape and lightness. The tool triangulated both maps and used texture mapping techniques to enable anti-aliased warps.
- Tool was optimized for human-in-the-loop operation and is used frequently in optometry labs at both UC Berkeley and University of Washington.

PUBLICATIONS

1. **Rishi Upadhyay**, Howard Zhang, Ayush Agrawal, Yunhao Ba, Alex Wong, Celso M de Melo, Achuta Kadambi. "WorldBench: Benchmarking Physical Understanding of World Models by Isolating Physics Concepts". *Under Submission*. (2025).
2. **Rishi Upadhyay**, Pascal Paysan, Supratik Bose, William Delery, Yunzheng Zhu, William Hsu, Daniel Low, Stefan Scheib, Achuta Kadambi, Ricky R. Savjani. "Learning Respiratory Dynamics in Fast Helical Free Breathing CT Imaging for Radiotherapy Planning". *Under Submission*. (2025).
3. Changwoo Jeon, **Rishi Upadhyay**, Achuta Kadambi. "MoCA3D: Monocular 3D Bounding Box Prediction in the Image Plane". *ECCV*. (2026).
4. Jiayi Yu*, Aki Takahashi*, Min-Ho Kim*, **Rishi Upadhyay***, Howard Zhang*, Bo Liu, Xintong Yuan, Thomas Hymel, Tianyu Wang, Kaiyan Liang, Keyue Liang, Haoyang Wu, Dingyi Zhao, Jung Tae Kim, Achuta Kadambi, Yuzhang Li. "Battery smart sensing via a virtual reference electrode". *Nature Communications*. (2025).
5. Haolin Xiong*, Sairisheek Muttukuru*, **Rishi Upadhyay**, Pradyumna Chari, Achuta Kadambi. "SparseGS: Real-Time 360° Sparse View Synthesis using Gaussian Splatting". *International Conference on 3D Vision (3DV)* (2025).
6. **Rishi Upadhyay**, Howard Zhang, Yunhao Ba, Ethan Yang, Blake Gella, Sicheng Jiang, Alex Wong, Achuta Kadambi. "Enhancing Diffusion Models with 3D Perspective Geometry Constraints". *ACM TOG (SIGGRAPH Asia)*. (2023).
7. Blake Gella*, Howard Zhang*, **Rishi Upadhyay**, Tiffany Chang, Nathan Wei, Matthew Waliman, Yunhao Bao, Celso de Melo, Alex Wong, Achuta Kadambi. "WeatherProof: Leveraging Language Guidance for Semantic Segmentation in Adverse Weather". *arXiv Preprint arXiv:2403.14874* (2024).
8. Vadim Ezhov*, Hyoungeob Park*, Zhaoyang Zhang*, **Rishi Upadhyay**, Howard Zhang, Chethan Chinder Chandrappa, Achuta Kadambi, Yunhao Ba, Julie Dorsey, Alex Wong. "All-day Depth Completion" *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. (2024).
9. Vimal Prabhu Pandiyan, Sierra Schleufer, Emily Slezak, James Fong, **Rishi Upadhyay**, Austin Roorda, Ren Ng, Ramkumar Sabesan. "Characterizing Cone Spectral Classification by Optoretinography". *Biomedical Optical Express* 13, 6574-6594 (2022).

TEACHING

EE102: Signals and Systems, UCLA

Spring 2025

EE149: Foundations of Computer Vision, UCLA

Winter 2024, Fall 2024

EE239AS: Computational Imaging, UCLA

Fall 2023

CS184: Computer Graphics & Imaging, UC Berkeley

Spring 2020, 2021, 2022

SKILLS

Languages: Python, C++, Triton, CUDA, HIP

Libraries: Pytorch, MONAI, Pytorch Lightning, vLLM-Omni, ROCm, Pytorch3D, timm, OpenCV, Slurm